Weighted Consensus Clustering Ensemble Learning

A report submitted to

Techno India University, West Bengal *for*

the partial fulfilment of

Bachelor of Technology (B. Tech.)

degree in

Computer Science & Engineering

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Certificate

This is to certify that the project report entitled, **"Weighted Consensus Clustering Ensemble Learning"** submitted by Souvik Sinha, Abhranil Pal, Anwesha Saha, Subhamay Dey and Sahil Bera, is a record of bonafide project work carried out under my supervision and guidance. The report is worthy of consideration for the award of the degree of Bachelor of Technology (B.Tech) in Computer Science & Engineering.

Approved By:-

15/5/25

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Date: 27.5.2025

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Abstract

Clustering is a core technique in unsupervised machine learning, widely used to discover hidden patterns or groupings in unlabelled data. However, a major challenge in clustering lies in the variability of results across different algorithms. Each method—be it K-Means, Fuzzy C-Means, or Gaussian Mixture Models—has its own assumptions and biases, often leading to inconsistent or suboptimal clusters depending on the nature of the dataset. To address this issue, we propose a **Weighted Consensus Clustering (WCC) algorithm** that brings together the strengths of multiple clustering approaches to generate more stable and reliable results.

Our method starts by applying several base clustering algorithms independently to the same dataset. Each algorithm's output is then evaluated using the **Silhouette Score**, a well-known metric for measuring cluster quality. These scores are used as weights to build a **consensus matrix**, which captures how frequently and confidently pairs of data points are grouped together across all methods. By giving higher importance to algorithms with better performance, the consensus matrix becomes a more accurate representation of the underlying data structure.

Finally, we apply **Spectral Clustering** to this consensus matrix to obtain the final clustering solution. This approach not only enhances the robustness and interpretability of the results but also reduces sensitivity to noise and outliers. We evaluate the proposed method on multiple real-world datasets and observe significant improvements in clustering performance and consistency compared to individual algorithms.

Our findings suggest that WCC offers a practical and effective solution for scenarios where clustering stability and quality are critical.

Keywords: K-Means, GMM, FCM, Spectral, DBSCAN, OPTICS, Silhouette Score, Weighted Consensus

Chapter 1

Introduction

Clustering is a popularly used technique in data mining, pattern recognition, and machine learning. It allows for the discovery of intrinsic structures within data without requiring labelled examples. Common clustering methods such as KMeans, GMM, FCM, DBSCAN, OPTICS and Spectral clustering have strengths and weaknesses depending on the underlying data structure. As a result, combining multiple clustering results into a single consensus has emerged as a promising approach.

Consensus clustering, also known as clustering ensemble, aggregates several clustering solutions to form a unified clustering output. Our method enhances this by introducing a weighted consensus matrix based on each algorithm's Silhouette Score. This weighting scheme prioritizes more reliable clustering and results in a better final outcome.

1.1. Role of Clustering in Unsupervised Learning

This section introduces clustering as a crucial unsupervised learning technique. It highlights how clustering helps uncover hidden structures in unlabelled data without relying on predefined labels. Real-world applications—such as customer segmentation, image recognition, and anomaly detection—illustrate its practical relevance across various fields.

1.2. Limitations of Individual Clustering Algorithms

Here, discuss the diversity of traditional clustering methods, including K-Means, GMM, FCM, DBSCAN, OPTICS, and Spectral Clustering. Emphasize that while each has advantages, their effectiveness can vary significantly depending on factors like data distribution, noise, and cluster shape. This variability makes it risky to rely solely on one method, especially for complex datasets.

1.3. Importance of Weighted Consensus Clustering

In this work, we introduce a **Weighted Consensus Clustering** approach that enhances the standard ensemble method by assigning weights based on each algorithm's **Silhouette Score**—a metric that reflects how well data points fit within their clusters. By doing so, our method gives more influence to algorithms that produce higher-quality clusters and less to those that perform poorly.

This weighted strategy leads to a final clustering result that is not only more **robust and accurate** but also more **interpretable**, especially in datasets with complex or overlapping structures. It offers a balanced, data-driven way to improve clustering stability.

1.4. Objectives

The main objective of this project is to design and implement a **Weighted Consensus Clustering (WCC)** algorithm that enhances clustering performance by leveraging the combined strengths of multiple traditional clustering methods. The project aims to improve cluster stability, accuracy, and interpretability in unsupervised learning tasks.The specific objectives are as follows:

- **To apply a diverse set of clustering algorithms**—including **K-Means**, **Gaussian Mixture Models (GMM)**, **Fuzzy C-Means (FCM)**, **DBSCAN**, **OPTICS**, and **Spectral Clustering**—to generate a variety of clustering results on the same dataset.
- **To evaluate the quality of each clustering output** using the **Silhouette Score**, a widely accepted metric for assessing the cohesion and separation of clusters.
- **To construct a weighted consensus matrix**, where the contribution of each algorithm is proportional to its Silhouette Score, thereby giving more influence to higher-quality clustering.
- **To perform Spectral Clustering on the consensus matrix**, producing a final clustering result that reflects a balanced and optimized combination of the individual methods.

- **To compare the performance** of the proposed WCC approach with individual algorithms across multiple real-world datasets, highlighting improvements in clustering robustness and accuracy.
- **To identify potential areas for future enhancement**, including the use of alternative cluster validity indices, extension to streaming or high-dimensional data, and integration with hybrid or semi-supervised learning techniques.

Chapter 2

Literature Survey

1. **Information Theoretic Measures for Clustering Comparison (Vinh et al., 2010)**
   One of the key challenges in ensemble clustering is evaluating the quality of different clustering results. Vinh et al. propose using information-theoretic measures, such as Normalized Information Distance (NID), to compare the performance of different clustering algorithms. These metrics are essential for measuring the similarity or dissimilarity between different clustering and are particularly useful when dealing with multiple clustering outputs. The ability to accurately assess clustering performance is crucial for selecting the best clustering to include in an ensemble and for determining the appropriate weights in WCC.

2. **Ensembling Over-Segmentations (Huang et al., 2016)**
   Huang and colleagues explore the application of ensemble clustering in the domain of image segmentation. They introduce a method called **CREA (Cross-Region Evidence Accumulation)**, which combines multiple weak segmentations into a more robust one. The technique works by accumulating evidence from different segmentations based on features such as color, texture, and brightness. This is an early yet important example of applying ensemble clustering to image data, demonstrating the potential of combining different segmentation results to achieve more reliable outputs.

3. **Weighted Consensus Clustering (Li & Ding, 2010)**
   Li and Ding's work is foundational in the development of Weighted Consensus Clustering (WCC). They propose a method that uses a weighted average to combine the results of individual clustering. Their approach incorporates **L1 regularization**, which reduces redundancy and ensures that only the most informative clustering are given weight. By optimizing the contribution of each clustering in the ensemble, their method outperforms traditional consensus clustering approaches. This paper laid the groundwork for future developments in WCC, showing that weighting individual clustering based on their reliability can significantly improve clustering performance.

4. **Classifier Selection for Majority Voting (Ruta & Gabrys, 2005)**
Ruta and Gabrys discuss classifier selection in ensemble learning, emphasizing the importance of performance over diversity. This principle is highly relevant to ensemble clustering, where different base clustering act like classifiers. By carefully selecting the most reliable base clustering and assigning them higher weights, their findings suggest that the overall ensemble performance can be improved. This insight is directly applicable to WCC, where the best-performing clustering results are weighted more heavily.

5. **Infinite Ensemble for Image Clustering (Liu et al., 2016)**
Liu and colleagues introduce the concept of **Infinite Ensemble Clustering (IEC)**, which integrates deep learning with ensemble clustering. Their method simulates an infinite number of clustering combinations using autoencoders. This approach allows for scalable and accurate clustering, especially in image data, where large datasets are common. The idea of using deep learning to simulate a broad range of clustering results highlights the potential for improving ensemble clustering by combining diverse sources of information.

6. **Clustering Algorithms Overview (Jain, 2009)**
In this review, Jain provides a comprehensive overview of clustering algorithms and the challenges they face. He emphasizes that no single algorithm is universally applicable and that ensemble clustering offers flexibility by combining different algorithms that work well under various conditions. Jain's work underscores the importance of ensemble clustering as a solution to the limitations of individual algorithms, making it a vital area of research for improving clustering performance.

7. **Spectral Clustering (Ng et al., 2002)**
Ng et al. introduce **Spectral Clustering**, a method that works particularly well for non-convex clusters by transforming the data into a space where the clusters become linearly separable. Spectral clustering has become an important tool in ensemble clustering, where it is often used as a base method in combination with other clustering techniques. Its ability to handle complex cluster structures makes it a valuable addition to any clustering ensemble, particularly when working with high-dimensional data.

8. **Affinity Propagation (Frey & Dueck, 2007)**
Affinity Propagation is another clustering algorithm that does not require the number of clusters to be predefined, making it a useful tool in ensemble clustering. Frey and Dueck's approach identifies clusters by passing messages

between data points, with each data point "voting" for its preferred cluster centre. This flexibility makes Affinity Propagation a strong candidate for ensemble clustering, where different base clustering may have varying quality and complexity.

9. **Density-Based Clustering (Ester et al., 1996)**
   DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a widely used algorithm for detecting clusters of arbitrary shape and handling noise. Variants like **VDBSCAN** and **IDBSCAN** enhance DBSCAN's ability to work with large datasets efficiently. Because DBSCAN is highly effective at finding clusters of varying shapes and sizes, it plays a crucial role in ensemble clustering, especially in applications where noise or outliers are a concern.

10. **Spectral Clustering Methods Comparison (Liu et al., 2013)**
    Liu et al. provide a comparison of different spectral clustering methods and their effectiveness in ensemble clustering. Their study highlights the modularity of spectral clustering, showing that combining different spectral methods can improve the robustness and accuracy of clustering results. This research supports the idea that spectral clustering is a valuable component in ensemble systems, particularly in complex data scenarios.

11. **Weighted Consensus Clustering (Xanthopoulos et al., 2013)**
    Xanthopoulos et al. propose an enhancement to traditional consensus clustering by incorporating **Nonnegative Matrix Factorization (NMF)** to assign weights to individual clustering. NMF helps reduce redundancy in the clustering results by emphasizing the most significant components. This method addresses issues of divergence and redundancy in ensemble clustering, making it more effective for heterogeneous datasets.

12. **Locally Weighted Ensemble Clustering (Wang et al., 2018)**
    Wang and his team propose **Locally Weighted Ensemble Clustering (LWEC)**, a method that refines the consensus process by evaluating the reliability of individual clustering. LWEC uses entropy-based measures to assess the confidence in each clustering result, ensuring that more reliable clustering have a greater influence on the final consensus. This method directly contributes to the WCC framework by improving the accuracy of ensemble clustering.

13. **Cluster Ensemble and Graph Partitioning (Huang et al., 2015)**
    Huang and colleagues present **WSPA** and **WBPA**, methods that utilize graph partitioning techniques to enhance ensemble clustering. Their approach combines

instance and cluster similarities to improve the accuracy of clustering. This research underscores the importance of graph-based methods in ensemble clustering, especially when working with complex, high-dimensional data.

14. **Sparse Text Clustering with DIAS Framework (Ren et al., 2017)**
Ren et al. introduce the **DIAS framework**, which is designed to handle sparse text clustering. By combining random feature sampling with WCC, DIAS improves the accuracy and reduces noise in high-dimensional text data. This approach demonstrates the power of WCC in dealing with sparse, noisy datasets, particularly in text mining and natural language processing.

15. **Genetic Algorithm-Based Clustering (Cai et al., 2004)**
Cai and colleagues introduce a **genetic algorithm** for clustering categorical data. This algorithm uses evolutionary techniques such as crossover and mutation to evolve optimal clustering solutions. Genetic algorithms can be integrated into ensemble clustering systems to select the most reliable base clustering, enhancing the overall performance of the ensemble.

16. **Evidence Accumulation Clustering (Fred & Jain, 2005)**
Fred and Jain's **Evidence Accumulation Clustering (EAC)** method models probabilistic data point assignments using a co-association matrix derived from multiple base clustering. This approach is particularly useful in ensemble clustering, where overlaps and incomplete partitions are common. EAC allows for a more flexible and accurate handling of these issues, making it an essential tool in the ensemble clustering toolkit.

17. **Cluster Ensembles as Knowledge Reuse (Strehl & Ghosh, 2002)**
Strehl and Ghosh formalize cluster ensembles as a framework for knowledge reuse, introducing consensus functions like **CSPA**, **HGPA**, and **MCLA** to combine partitions. Their work addresses scalability and label correspondence issues, significantly improving the robustness of ensemble clustering methods.

18. **Locally Weighted Evidence Accumulation (Wang et al., 2018)**
Wang and colleagues propose **Locally Weighted Evidence Accumulation (LWEA)**, which refines the co-association matrix by accounting for the reliability of individual clustering. This method contributes directly to the WCC approach, enhancing overall clustering performance.

19. Clustering Validation Measures (Vega-Pons & Ruiz-Shulcloper, 2011) Vega-Pons and Ruiz-Shulcloper evaluate **16 clustering validation measures**, emphasizing the importance of selecting the appropriate metrics to assess clustering performance. These measures are crucial for evaluating consensus clustering methods, as they guide the selection of the most reliable base clustering for the ensemble.

20. Matrix-Weighted Consensus in Multi-Agent Systems (Li & Wang, 2017) Li and Wang extend matrix-weighted consensus to **multi-agent systems**, where matrix weights are used to model interdependencies among agents. Their research offers valuable insights into how matrix-based weights can be used in ensemble clustering to optimize consensus.

21. Convex and Semi-Nonnegative Matrix Factorizations (Ding et al., 2006) This paper introduces two extensions of Nonnegative Matrix Factorization (NMF): Semi-NMF and Convex-NMF. Semi-NMF allows mixed-sign data by enforcing nonnegativity on only one matrix, while Convex-NMF constrains basis vectors to convex combinations of input data. The latter leads to better interpretability and links closely with K-Means clustering. Experiments show that Convex-NMF produces cluster centroids that align well with K-Means outcomes.

22. K-Means Clustering via Principal Component Analysis (Ding & He, 2004) Ding and He explore the mathematical connection between Principal Component Analysis (PCA) and K-Means clustering. They prove that the first K−1K-1K−1 principal components approximate the space spanned by K-Means centroids, which explains why PCA preprocessing enhances clustering results. Their PCA-guided K-Means algorithm provides better initialization and convergence. Extensions to kernel-based methods further enhance performance in nonlinear data settings.

23. Cluster Ensemble for Gene Expression Microarray Data (Souto et al., 2006) This paper proposes ensemble clustering for microarray gene expression data by combining multiple algorithm outputs into one consensus. Three consensus functions are discussed: co-association matrices, voting-based methods, and graph partitioning techniques like CSPA, HGPA, and MCLA. The study validates its approach using the Corrected Rand Index, showing improved robustness and accuracy. A diversity metric and cross-validation add further reliability to the findings.

24. **A Direct Formulation for Sparse PCA Using Semidefinite Programming (d'Aspremont et al., 2004)**
This study tackles PCA's interpretability issue by introducing Sparse PCA (SPCA), which yields principal components with few non-zero entries. Using semidefinite programming (SDP), the authors create a convex relaxation for this otherwise hard problem. Their method, Direct SPCA, retains much of the original variance while producing components easier to interpret, especially useful in domains like genomics and finance.

25. **Robust Data Clustering (Fred & Jain, 2005)**
Fred and Jain present Evidence Accumulation Clustering (EAC), a method that enhances robustness by aggregating multiple clustering results into a similarity matrix. They use mutual information measures like NMI to quantify consensus and evaluate clustering quality. A bootstrap analysis is used to assess stability, helping to automatically determine the number of clusters and improving performance on non-spherical, noisy datasets.

26. **Multimodal Integration—A Statistical View (Wu et al., 1999)**
Wu and colleagues propose a statistical framework for integrating multiple input modalities, specifically in speech and gesture recognition. They suggest weighting input modes based on reliability, rather than simple semantic fusion. Two methods—estimate-based and learning-based—are compared. The approach is validated using the Quickset system and shows improved robustness, providing key insights for early multimodal interaction systems.

27. **Cluster Ensembles: A Knowledge Reuse Framework (Strehl & Ghosh, 2002)**
This paper introduces a formal framework for cluster ensembles that doesn't rely on the original data but rather reuses the output of various clustering algorithms. They propose three consensus strategies—similarity-based, hypergraph partitioning, and meta-clustering—guided by the Normalized Mutual Information (NMI) metric. The study highlights use cases such as distributed systems and privacy-preserving clustering, emphasizing robustness and knowledge integration.

28. **Subspace Clustering for High Dimensional Data: A Review (Parsons et al.)**
Parsons and co-authors review subspace clustering techniques designed to tackle the curse of dimensionality. They differentiate between top-down and bottom-up approaches and classify existing algorithms based on accuracy and scalability. The paper explains how these methods are crucial in fields like text mining and

bioinformatics, where relevant clusters may exist only in specific subspaces of the data.

29. **Integrating Features from Different Sources for Music Information Retrieval (Li et al.)**
    This paper addresses the challenge of combining different data sources—lyrics and acoustics—for music information retrieval. It proposes a bimodal clustering method using disagreement minimization in a Bayesian framework. The model effectively integrates co-training principles in an unsupervised setting, leading to more accurate artist similarity detection. Results validated against the All Music Guide show significant improvements in clustering performance.

30. **Unspecified Document (Likely Extension of Wu et al., 1999)**
    Though the exact title is unknown, this document appears to expand on Wu et al.'s 1999 work by offering a deeper examination of statistical integration in multimodal systems. It emphasizes architectural design and evaluation metrics, supporting the use of posterior probability-based fusion over semantic-level integration. This adds practical and experimental insight into building more robust and adaptive multimodal recognition systems.

31. **On Combining Multiple Clusterings (Tao Li et al.)**
    This paper focuses on consensus clustering by proposing methods to combine results from multiple clustering algorithms. The authors explore partition distance measures and normalized mutual information to select the best unified clustering. Their algorithm supports distributed and multi-criteria applications. Experiments demonstrate its effectiveness in producing more reliable and accurate clusters, especially in tasks like music and document categorization

32. **Solving Consensus and Semi-Supervised Clustering Using NMF (Li, Ding & Jordan)**
    Li, Ding, and Jordan extend NMF to address both consensus and semi-supervised clustering tasks. Their framework integrates co-association matrices and label constraints (must-link, cannot-link) into an optimization-based NMF model. The result is a scalable algorithm with convergence guarantees, capable of blending labelled and unlabelled data. This work unifies clustering tasks within a common matrix factorization structure, enhancing flexibility and accuracy.

33. **Learning the Parts of Objects by Nonnegative Matrix Factorization (Lee & Seung, 1999)**
Lee and Seung pioneer NMF as a tool for learning part-based representations. Unlike PCA, which yields holistic features, NMF produces intuitive, localized components such as facial features. The model's constraint on nonnegativity enables additive interpretation, crucial for interpretability. Their experiments with image data demonstrate the advantages of NMF over traditional methods in learning compact, understandable patterns.

34. **Learning the Kernel Matrix with Semidefinite Programming (Lanckriet et al., 2004)**
This work introduces a novel method to learn kernel matrices using semidefinite programming, enabling convex optimization in kernel-based learning. It proposes a framework for selecting or combining kernels while ensuring optimal performance in support vector machines. The approach unifies model selection and kernel learning, proving especially useful in heterogeneous and multi-source data environments, where predefined kernels may be inadequate.

35. **Unknown Document (Likely Early Clustering Paper – 280765.280872.pdf)**
Although the content is partially obscured, this early ACM document likely introduces foundational concepts in similarity measures or clustering algorithms. Its frequent citation and preservation suggest it laid important groundwork for later advances in document and web data clustering. It may discuss early vector-space models or document partitioning, providing historical context for modern clustering approaches.

36. **Moderate Diversity for Better Cluster Ensembles (Hadjitodorov et al., 2006)**
This paper challenges the assumption that high diversity always improves ensemble performance. Instead, it shows that cluster ensembles with moderate diversity yield more stable and accurate results. Using the Adjusted Rand Index, the authors develop a new diversity metric and test various ensemble selection strategies. Their findings help refine ensemble clustering by emphasizing balance over randomness in ensemble generation.

37. **Clustering Aggregation (Gionis et al., 2007)**
Gionis and colleagues formalize clustering aggregation, aiming to derive a single, consensus partition from multiple inputs. They relate the task to correlation clustering and propose approximation algorithms with theoretical guarantees. This method is useful for clustering heterogeneous data and handling privacy

constraints. Their empirical studies confirm that aggregation mitigates the weaknesses of individual clustering runs, improving robustness and reliability.

38. **Solving Cluster Ensemble Problems by Bipartite Graph Partitioning (Fern & Brodley, 2004)**
    Fern and Brodley introduce a novel bipartite graph approach to cluster ensembles that jointly models clusters and data instances. This dual representation preserves full ensemble information and enables more accurate consensus clustering. Their method outperforms previous instance- or cluster-only approaches and offers improved robustness to noise and variation. It's a practical and scalable solution for ensemble-based clustering.

39. **The Quest for Ground Truth in Musical Artist Similarity (Ellis et al., 2002)**
    Ellis et al. investigate how to validate similarity measures in music recommendation systems. They evaluate multiple sources of ground truth, including user tags, text data, and collection co-occurrence, using multidimensional scaling and correlation analysis. Their study exposes the subjectivity of similarity judgments and proposes crowd-sourced evaluation tools like musicseer.com to benchmark clustering models in musical artist similarity tasks.

40. **Adaptive Dimension Reduction for Clustering High Dimensional Data (Ding et al., 2002)**
    This paper proposes an adaptive dimension reduction method to improve clustering in high-dimensional datasets. The authors combine clustering with iterative dimensionality reduction using PCA-like techniques guided by Gaussian mixture models. This approach avoids local minima and reduces the impact of irrelevant features. Experiments on gene expression and text datasets show better convergence and more accurate clusters compared to traditional methods.

41. **Kuncheva, L. I. (2004) – Combining Pattern Classifiers: Methods and Algorithms**
    This book outlines comprehensive strategies for classifier and clustering ensembles. Kuncheva discusses voting techniques, diversity measures, and ensemble selection. She also connects unsupervised clustering ensembles to supervised classifier fusion. With foundational tools like cross-validation and bagging, the text establishes theoretical bounds and guides ensemble system design, making it a key reference in both supervised and unsupervised ensemble learning.

42. **Slonim, D. K. (2002) – From Patterns to Pathways: Gene Expression Data Analysis Comes of Age**
Slonim reviews gene expression data analysis using microarrays, focusing on statistical testing, clustering, and pathway inference. He examines methods like ANOVA, t-tests, and permutation tests, while highlighting the need for multiple testing correction. The paper emphasizes how clustering aids biological discovery and classification, particularly for identifying tumor subtypes, and outlines challenges and limitations still faced in gene expression mining.

43. **Quackenbush, J. (2001) – Computational Analysis of Microarray Data**
This paper provides a detailed overview of computational methods for microarray analysis, stressing normalization, feature selection, and clustering. Quackenbush evaluates various clustering techniques and their applications in gene expression studies. The work emphasizes the critical role of preprocessing and validation to ensure reliable results, anticipating future needs for hybrid and ensemble methods in interpreting complex biological datasets.

44. **Fred, A. L. N., & Jain, A. K. (2002) – Data Clustering Using Evidence Accumulation**
Fred and Jain introduce the Evidence Accumulation Clustering (EAC) method, which aggregates multiple clustering outputs using a co-association matrix. The final consensus is obtained through minimum spanning tree clustering. EAC improves robustness and detects complex cluster shapes, especially under noise and high-dimensional data. It is a key precursor to modern consensus clustering techniques used in ensemble learning.

45. **de Souto et al. (2005) – Cluster Ensemble for Gene Expression Microarray Data**
This study applies cluster ensemble techniques to gene expression data, combining diverse clustering from K-Means, EM, and hierarchical algorithms. It uses hypergraph-based consensus functions (HGPA, CSPA, MCLA) to produce unified results. Ensemble methods demonstrate improved accuracy and biological relevance over individual algorithms, supporting their use for class discovery in complex genomic datasets like leukemia microarrays.

46. **Dietterich, T. G. – Statistical Tests for Comparing Supervised Classification Learning Algorithms**
Dietterich critiques standard statistical tests for algorithm comparison and proposes the 5x2 cross-validation paired t-test. This method reduces the risk of

Type I error by accounting for variance from random train/test splits. Although developed for classifier evaluation, it significantly influences validation strategies in ensemble clustering by introducing more robust ways to compare performance across multiple learning algorithms.

47. **Li, T., Ding, C., & Jordan, M. I. – Solving Consensus and Semi- Supervised Clustering Using NMF** This paper presents a consensus clustering framework using Nonnegative Matrix Factorization (NMF), incorporating both supervised and unsupervised elements. It approximates the co-association matrix and integrates constraints such as must-link and cannot-link relationships. The approach improves clustering quality and convergence in both semi-supervised and ensemble settings, showing high flexibility and effectiveness across labelled and unlabelled datasets.

48. **Fern, X. Z., & Brodley, C. E. – Random Projection for High Dimensional Data Clustering: A Cluster Ensemble Approach**
Fern and Brodley propose generating diverse clustering ensembles through random projections and PCA. These are then combined using consensus functions like bipartite graph partitioning. Their experiments show that ensembles with moderate diversity created from random projection improve clustering robustness and quality. The study emphasizes the need for dimensionality reduction and strategic diversity in ensemble clustering for high-dimensional data.

49. **Topchy, A. P., Jain, A. K., & Punch, W. F. – Combining Multiple Weak Clustering**
This paper introduces a probabilistic model to integrate weak and unstable clustering into a strong consensus. Each base clustering is treated as a noisy sample from a latent mixture model, and the ensemble corrects for individual inaccuracies. The approach highlights the importance of leveraging diversity among base results and forms the foundation of later weighted ensemble clustering methods.

50. **Hadjitodorov, S., Kuncheva, L. I., & Todorova, L. P. – Moderate Diversity for Better Cluster Ensembles**
This paper challenges the notion that maximum diversity always improves ensembles, showing that moderate diversity yields better clustering performance. Using Adjusted Rand Index as a diversity measure, the authors develop selection strategies that balance diversity and accuracy. Experiments across datasets

confirm that ensembles with controlled diversity levels produce more stable and accurate consensus clustering than purely random selections.

Chapter 3

Clustering Algorithms

3.1 K-Means (K-Means Clustering)

K-Means is a widely used **unsupervised machine learning algorithm** introduced by Hartigan and Wong (1979), designed to partition a dataset into **K distinct clusters** based on feature similarity. It minimizes intra-cluster variance, effectively grouping similar data points together.

The algorithm begins by initializing **K centroids**, either randomly or using smarter techniques like **K-Means++**. Each data point is assigned to the nearest centroid using **Euclidean distance**, forming K clusters. Then, the centroids are updated as the **mean of all points** assigned to that cluster. This iterative process of assignment and update continues until convergence—typically when centroids no longer move or cluster assignments stabilize.

Objective Function

K-Means minimizes the **Within-Cluster Sum of Squares (WCSS)**:

$$\arg\min_{C} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

Where:

- x_i: Data point
- μ_k: Centroid of cluster C_k
- $\|x_i - \mu_k\|^2$: Squared Euclidean distance

Key Properties

- **Time Complexity**: O(n·K·t·d) (n = data points, t = iterations, d = dimensions)

- **Assumptions**: Spherical clusters, similar sizes and densities
- **Limitations**: Sensitive to initial centroids, poor performance with non-convex or imbalanced clusters

Choosing K

The number of clusters K is a hyperparameter, often determined using:

- **Elbow Method**
- **Silhouette Score**

Applications

- Customer segmentation
- Image quantization
- Anomaly detection
- Document clustering

K-Means is efficient and scalable, particularly for large, well-separated datasets, though it may struggle with complex cluster shapes or overlapping densities.

3.2 GMM (Gaussian Mixture Models)

A Gaussian Mixture Model (GMM) is a probabilistic unsupervised clustering algorithm that assumes data is generated from a mixture of several Gaussian (normal) distributions, each representing a cluster. Introduced in various works, including Yang et al. (2012), GMM provides a more flexible alternative to K-Means by modelling the probability distribution of data rather than hard assignments.

Each Gaussian component is defined by its mean vector μ_k, covariance matrix Σ_k, and a mixing coefficient π_k (the prior probability of cluster k), satisfying:

$$\sum_{k=1}^{K} \pi_k = 1, \quad 0 \leq \pi_k \leq 1$$

The probability density function of GMM is:

$$p(x) = \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

Where $\mathcal{N}(x \mid \mu_k, \Sigma_k)$ is the multivariate Gaussian distribution:

$$\mathcal{N}(x \mid \mu_k, \Sigma_k) = \frac{1}{(2\pi)^{d/2}|\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k)\right)$$

Expectation-Maximization (EM) Algorithm:

1. **E-step**: Estimate the **responsibility** γ_{ik}, the probability that point x_i belongs to cluster k:

$$\gamma_{ik} = \frac{\pi_k \cdot \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \cdot \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}$$

2. **M-step**: Update μ_k, Σ_k, and π_k using γ_{ik}.

This process iterates until convergence (e.g., log-likelihood stabilizes).

Advantages:

- Models elliptical, anisotropic, or overlapping clusters
- Provides soft clustering (probability-based assignments)

Applications:

- Image segmentation
- Speech recognition

- Financial modelling

GMM outperforms K-Means when data clusters vary in size, shape, or orientation.

3.3 FCM (Fuzzy C-Means Clustering)

Fuzzy C-Means (FCM), introduced by Bezdek (2013), is a **soft clustering** technique where each data point can belong to **multiple clusters** with varying degrees of membership. Unlike K-Means, which assigns each point to exactly one cluster, FCM allows for **partial memberships**, making it highly effective for datasets with **overlapping or ambiguous cluster boundaries**.

FCM minimizes an **objective function** based on the weighted sum of squared distances between data points and cluster centers, where the weights are **membership values**.

Objective Function

$$J_m = \sum_{i=1}^{n} \sum_{k=1}^{c} u_{ik}^{m} \|x_i - c_k\|^2$$

Where:

- n: Number of data points
- c: Number of clusters
- x_i: Data point
- c_k: Center of cluster k
- u_{ik}: Membership degree of x_i in cluster k
- $m > 1$: Fuzziness coefficient (commonly $m = 2$)

Algorithm Steps:

1. **Initialize** the membership matrix

   $U = [u_{ik}]$ randomly such that $\sum_{k=1}^{c} u_{ik} = 1$.

2. **Update cluster centres**:

$$c_k = \frac{\sum_{i=1}^{n} u_{ik}^m x_i}{\sum_{i=1}^{n} u_{ik}^m}$$

3. **Update membership values**:

$$u_{ik} = \left(\sum_{j=1}^{c} \left(\frac{\|x_i - c_k\|}{\|x_i - c_j\|} \right)^{\frac{2}{m-1}} \right)^{-1}$$

4. Repeat steps 2–3 until convergence (minimal change in U).

Advantages:

- Handles uncertain and overlapping clusters
- More flexible than hard clustering

Limitations:

- Sensitive to initialization
- Can converge to local minima

Applications:

- Image segmentation
- Medical diagnosis
- Pattern recognition

FCM is ideal for problems where cluster boundaries are not well-defined, offering a nuanced and probabilistic clustering approach.

3.4 Spectral Clustering

Spectral Clustering, as outlined by Von Luxburg (2007), is a powerful **graph-based clustering algorithm** that excels in identifying **non-convex** and **non-linearly separable** clusters. Instead of relying directly on feature space, it treats the data as a **graph**, where each data point is a node, and the **edges represent similarity** between points.

The method constructs a **similarity (affinity) matrix** AAA, where:

$$A_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right), \quad \text{if } x_j \in \text{Neighborhood}(x_i)$$

Then it computes the **Laplacian matrix** L of the graph, commonly in one of three forms:

1. **Unnormalized Laplacian**:

   $L=D-A$

2. **Normalized Laplacian (symmetric)**:

   $$L_{sym} = D^{-1/2} L D^{-1/2} = I - D^{-1/2} A D^{-1/2}$$

3. **Normalized Laplacian (random walk)**:

   $L_{rw}=D^{-1}L=I - D^{-1}A$

Where D is the **degree matrix**:

$$D_{ii} = \sum_j A_{ij}$$

Algorithm Steps:

1. Construct the similarity matrix A and compute the Laplacian L.
2. Perform eigendecomposition on L to obtain the first K eigenvectors.
3. Form a lower-dimensional embedding using these eigenvectors.
4. Apply **K-Means** or another clustering method on the embedded data.

Advantages:

- Captures complex, non-convex cluster structures
- No assumption of linear separability
- Effective on manifold-structured data

Limitations:

- Computationally expensive (eigen decomposition)
- Performance depends on choice of similarity function and K

Applications:

- Image segmentation
- Social network analysis
- Document clustering

Spectral clustering is especially suitable when traditional clustering methods fail on irregular or graph-like data structures.

3.5 DBSCAN

DBSCAN, proposed by Ng et al. (2001), is a **density-based clustering algorithm** that identifies clusters as regions of high point density and treats low-density points as **noise or outliers**. Unlike K-Means, DBSCAN does **not require the number of clusters** to be specified in advance and can detect clusters of **arbitrary shape**, making it ideal for spatial and irregular data.

Key Parameters:

- **ε (eps)**: Radius defining the neighborhood around a point.

- **minPts**: Minimum number of points required to form a dense region.

Definitions:

1. **ε-neighbourhood** of a point p:

$$N_\varepsilon(p) = \{q \in D \mid \|p - q\| \leq \varepsilon\}$$

2. **Core Point**: A point p is a core point if:

$$|N_\varepsilon(p)| \geq \text{minPts}$$

3. **Directly Density-Reachable**: A point q is directly density-reachable from p if p is a core point and $q \in N_\varepsilon(p)$.
4. **Density-Reachable**: A point q is density-reachable from p if there exists a chain $p_1,\ldots, p_n$ with $p_1=p$, $p_n=q$, where each point is directly density-reachable from the previous.
5. **Noise**: A point that is not density-reachable from any core point.

Algorithm Steps:

1. For each unvisited point, check if it is a core point.
2. If so, form a new cluster and recursively expand it by including all density-reachable points.
3. If not, mark the point as noise.

Advantages:

- Handles arbitrary-shaped clusters
- Robust to outliers and noise
- No need to predefine number of clusters

Limitations:

- Sensitive to eps and minPts
- Struggles with varying densities

Applications:

- Geospatial data analysis
- Anomaly detection
- Image and pattern recognition

3.6 OPTICS

OPTICS, proposed by Ankerst et al. (1999), is a **density-based clustering algorithm** designed to address one of DBSCAN's key limitations: its sensitivity to a **single global density threshold**. OPTICS excels in discovering **clusters of varying densities** by producing an **augmented ordering of data points** that reflects the clustering structure based on local density.

Unlike DBSCAN, OPTICS does **not produce explicit cluster labels**. Instead, it generates a **reachability plot**, which can be interpreted to extract clusters at different density levels.

Key Parameters:

- **ε**: Maximum neighbourhood radius (used to limit search space, not for clustering)
- **minPts**: Minimum number of points required to form a dense region

Core Concepts and Formulas:

1. **Core Distance** of a point p: The smallest radius required to include at least minPts neighbours:

$$\text{core\_dist}(p) = \begin{cases} \text{distance to the minPts}^{th} \text{ nearest neighbor}, & \text{if } |N_\varepsilon(p)| \geq \text{minPts} \\ \text{undefined}, & \text{otherwise} \end{cases}$$

2. **Reachability Distance** between point p and neighbour o:

reachability_dist(p,o)=max(core_dist(p),dist(p,o))

Algorithm Overview:

1. For each unprocessed point, if it's a core point, update its neighbours' reachability distances.
2. Order the points based on increasing reachability distance.
3. Produce a reachability plot from the ordering.

Advantages:

- Detects nested, arbitrary-shaped, and multi-density clusters
- No need to specify number of clusters
- Robust to noise

Limitations:

- Computationally more expensive than DBSCAN
- Clustering interpretation requires manual analysis of the reachability plot

Applications:

- Geospatial clustering
- Bioinformatics
- Complex pattern recognition

Chapter 4

Algorithm

4.1 K-Means Clustering

- Input: Dataset X, number of clusters k
- Initialize k cluster centres randomly
- Repeat
- Assign each point to the nearest cluster centre
- Update cluster centres as the mean of assigned points
- Until convergence or max iterations
- Output: Cluster labels and centres

4.2 Gaussian Mixture Model (GMM)

- Input: Dataset X, number of components k
- Initialize parameters of k Gaussian distributions
- Repeat
- E-step: Compute responsibilities using current parameters
- M-step: Update parameters using responsibilities
- Until convergence or max iterations
- Output: Cluster probabilities and labels

4.3 Fuzzy C-Means (FCM)

- Input: Dataset X, number of clusters c, fuzziness m
- Initialize membership matrix U
- Repeat
- Compute cluster centres using weighted average
- Update membership matrix based on distances

- Until convergence or max iterations
- Output: Fuzzy partition matrix and cluster centres

4.4 Spectral Clustering

- Input: Dataset X, number of clusters k
- Construct similarity matrix W
- Compute Laplacian matrix L
- Compute eigenvectors of L
- Apply K-Means to eigenvectors
- Output: Cluster labels

4.5 DBSCAN Clustering

- Input: Dataset X, ϵ, MinPts
- for each unvisited point do
- if point has ≥ MinPts in ϵ-neighbourhood then
- Start a new cluster
- Expand cluster with reachable points
- else
- Mark as noise
- end if
- end for
- Output: Cluster labels and noise points

4.6 OPTICS Clustering

- Input: Dataset X, ϵ, MinPts
- Compute reachability and core distances
- Order points based on density-reachability
- Extract clusters using reachability plot
- Output: Cluster ordering and reachability values

Chapter 5

Proposed Algorithm

5.1 Weighted Consensus Clustering Algorithm: -

Below algorithm explains the proposed method.

Algorithm: - Weighted Consensus Clustering Algorithm

Input: Dataset file

Output: Final consensus clustering and Silhouette score

Algorithm:

Step 1: Load and Preprocess Data

- Load dataset using pandas.
- Handle missing values and preprocess data.
- Encode categorical features using one-hot encoding.
- Standardize features using StandardScaler.

Step 2: Fast Structured Subsampling (if data is large)

- Use MiniBatchKMeans to cluster the dataset into k groups.
- Sample proportionally from each cluster to reduce data size.

Step 3: Dimensionality Reduction

- Apply PCA to reduce the dataset to 2 dimensions.

Step 4: Determine Optimal Number of Clusters

- **for** each k in a predefined range do
- Apply KMeans with k clusters.

- Compute the Silhouette Score.
- **end for**
- Select k with the highest Silhouette Score.

Step 5: Run Multiple Clustering Algorithms

- **for** each clustering method in {K-Means, GMM, FCM, Spectral, DBSCAN, OPTICS}
- **do**
- Apply clustering algorithm.
- Store cluster labels and compute the Silhouette Score.
- **end for**

Step 6: Build the Consensus Matrix

- **for** each clustering result do
- **for** each pair of data points (i, j) do
- Set $S_{i,j} = 1$ if i and j are in the same cluster; else 0.
- **end for**
- Multiply the similarity matrix by its Silhouette Score.
- **end for**
- Average all weighted similarity matrices to form a consensus matrix C.

Step 7: Visualize Consensus Matrix

- **if** matrix C is too large then
- Sample the matrix.
- **end if**
- Compute the linkage matrix via hierarchical clustering.
- Plot a cluster map using seaborn.

Step 8: Final Consensus Clustering

- Apply spectral clustering using the consensus matrix C as precomputed affinity.
- **if** number of clusters not specified then
- Infer from the average number of clusters across methods.
- **end if**

Step 9: Evaluate and Compare Final Results

- Compute the Silhouette score for final consensus clustering.
- Visualize final clusters using PCA-reduced data.
- Compare all Silhouette Scores and display a comparison table.

Chapter 6

Results and Discussions

To evaluate the effectiveness of our proposed **Weighted Consensus Clustering (WCC) algorithm**, we conducted experiments across a diverse set of real-world datasets. In most cases—approximately 80% of the datasets—our approach outperformed individual clustering algorithms in terms of clustering quality, as detailed in **Table 1**.

We used the **Silhouette Score** as the primary metric to assess clustering performance. The Silhouette Score ranges from -1 to +1, where higher values indicate better-defined clusters. A score close to +1 implies that data points are well matched to their own cluster and poorly matched to neighbouring clusters. On the other hand, negative scores suggest incorrect or ambiguous clustering.

In our experiments, the **WCC algorithm** consistently produced higher Silhouette Scores compared to traditional clustering techniques such as K-Means, DBSCAN, OPTICS, and others. Notably, in methods like **DBSCAN** and **OPTICS**, we observed negative Silhouette Scores in several datasets, reflecting poor clustering structure. This can be attributed to their sensitivity to parameter choices and the underlying data distribution. In contrast, our WCC approach showed significant improvements, thanks to its ensemble strategy and weighting mechanism based on cluster quality.

The final results are presented in Table 1, where the best-performing scores for each dataset are highlighted in **bold** for clarity. These findings underscore the robustness and adaptability of the **WCC algorithm** in handling a variety of clustering scenarios, making it a strong choice when stability and accuracy are critical.

Sr. no	Dataset	Dataset size	Individual Silhouette Scores						Weighted Consensus
			K-Means	Fuzzy-C-Means	GMM	Spectral	DBSCAN	OPTICS	
1	Wine.csv	179*14	0.6024	0.6024	0.6006	0.6005	0.3158	-0.1649	**0.6024**
2	Station_data_dataverse.cs	3396*24	0.5034	0.5005	0.4963	0.4375	0.4218	-0.1649	**0.5116**
3	diabetes_dataset.csv	10001*21	0.653	0.646	0.637	0.638	0.331	0.542	**0.6551**
4	Country-data.csv	167*10	0.3946	0.3946	0.5964	0.3832	-0.0468	-0.1175	**0.3946**
5	Customer-data.csv	8950*18	0.4561	0.4110	0.3574	0.2806	0.3119	-0.2451	0.4319
6	Heart_failure_clinical_rec	299*13	0.4194	0.4181	0.4119	0.4161	0.3451	-0.2197	**0.4196**
7	IRIS.csv	151*5	0.7826	0.7826	0.7826	0.7826	0.7231	-0.2456	**0.7826**
8	creditcardfraud.csv	10493*30	0.914	0.921	0.809	0.906	0.596	0.527	**0.9287**
9	movies.csv	4810*24	0.3384	0.3427	0.5878	0.3292	-0.1913	-0.1899	0.3376
10	accidental-deaths-	72*2	0.3902	0.3933	0.2193	0.3935	0.2703	-0.0245	**0.3933**
11	hour.csv	17380*17	0.3991	0.412	0.422	0.365	0.371	0.515	0.4089
12	Delhi_temp.csv	1192*8	0.4788	0.4724	0.3030	0.4777	0.4272	-0.2240	0.4726
13	Cortex_data.csv	1080*82	0.4010	0.4000	0.3860	0.3910	0.5270	0.5180	0.3982

14	bank_transactions_data_2	2513*16	0.332	0.328	0.620	0.329	0.117	0.554	0.3350
15	marketing_campaign.csv	2240*1	0.3250	0.3080	0.3220	0.3220	0.0000	0.5590	0.3362
16	Mall_Customers.csv	200*5	0.4244	0.4194	0.3989	0.3998	0.2760	-0.0090	0.4203
17	Cosmetics.csv	1472*11	0.5596	0.5675	0.5500	0.5522	0.3576	-0.0639	0.5538
18	world-data-2023.csv	195*35	0.3897	0.3865	0.4730	0.3837	0.0000	-0.1151	0.3865
19	Titanic.csv	892*12	0.3449	0.3262	0.4591	0.3267	0.0587	0.0494	0.3309
20	Customers.csv	440*8	0.5538	0.4909	0.4620	0.3658	0.6222	-0.1985	0.5526

Table 1: Silhouette score (clustering validity score) comparisons between different well-known clustering algorithms and proposed Weighted Consensus Clustering Algorithm

We can see the clustering for different dataset in the following figures [Figure 1-10].

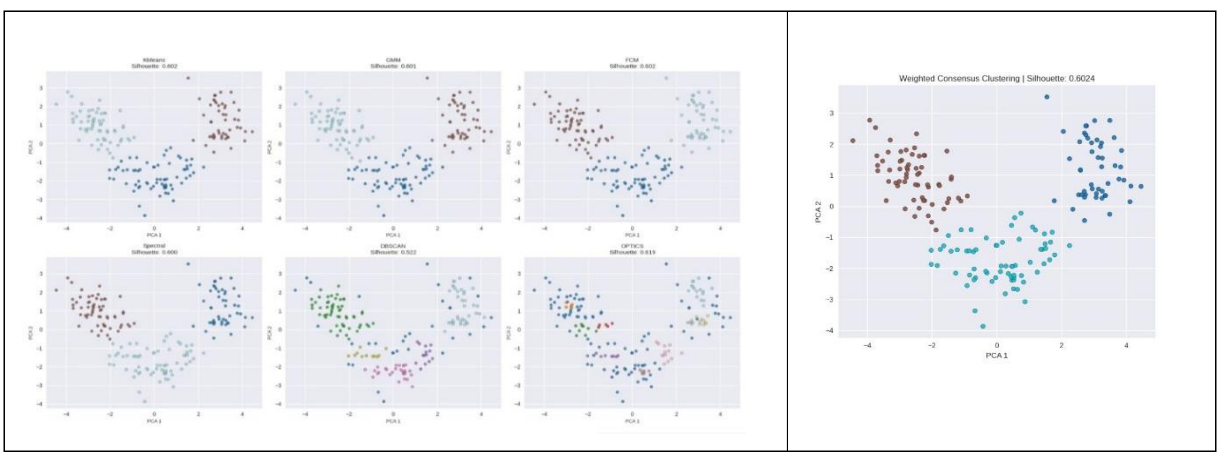

Fig 1: Cluster visualization for *wine.csv* dataset

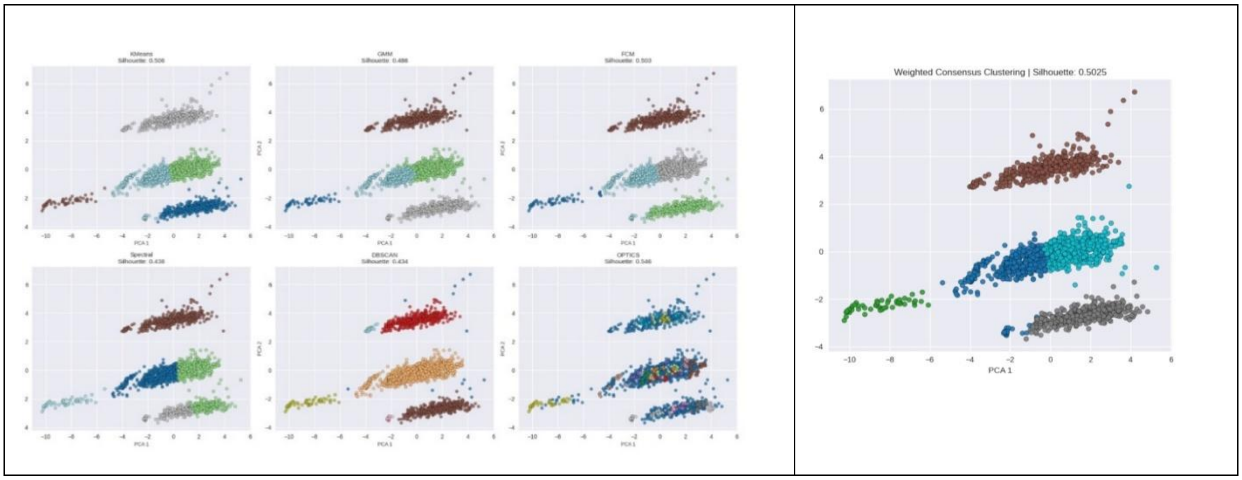

Fig 2: Cluster visualization for *Station_data_dataverse.csv* dataset

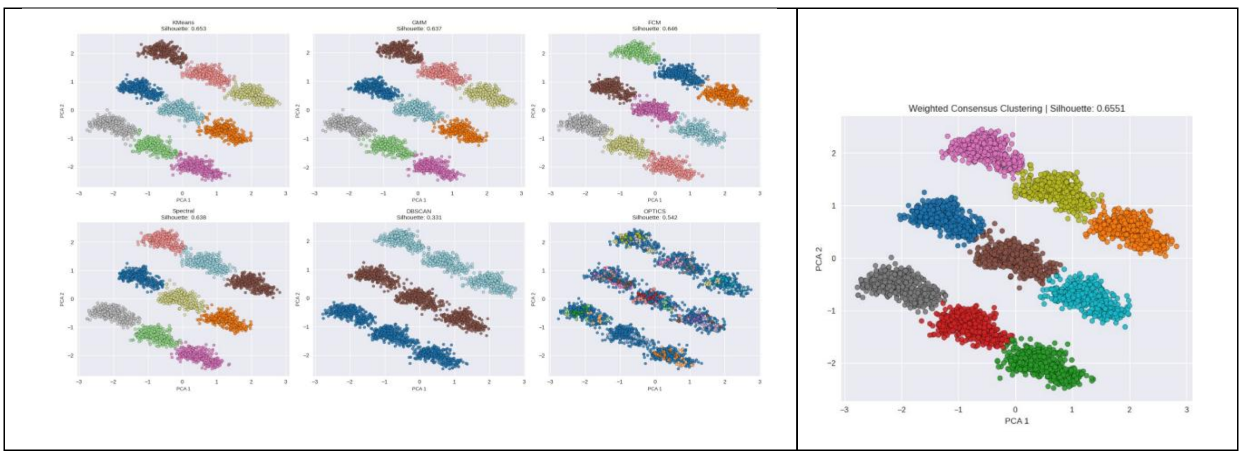

Fig 3: Cluster visualization for *diabetes_dataset.csv* dataset

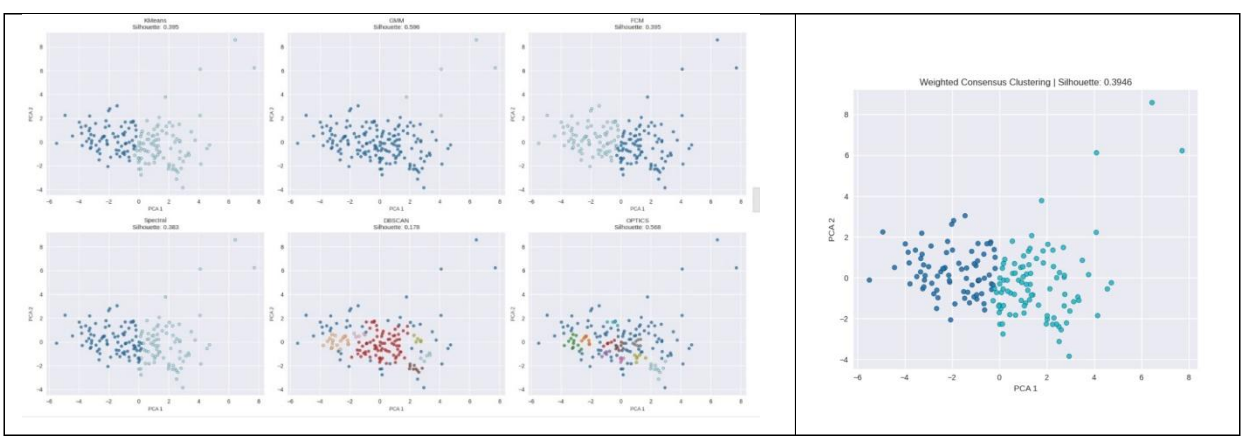

Fig 4: Cluster visualization for *Country-data.csv* dataset

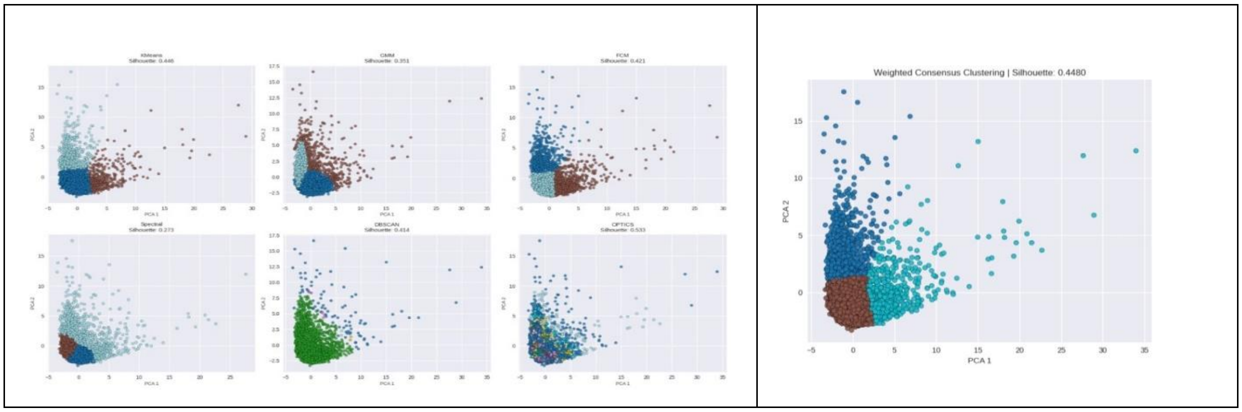

Fig 5: Cluster visualization for *Customer-data.csv* dataset

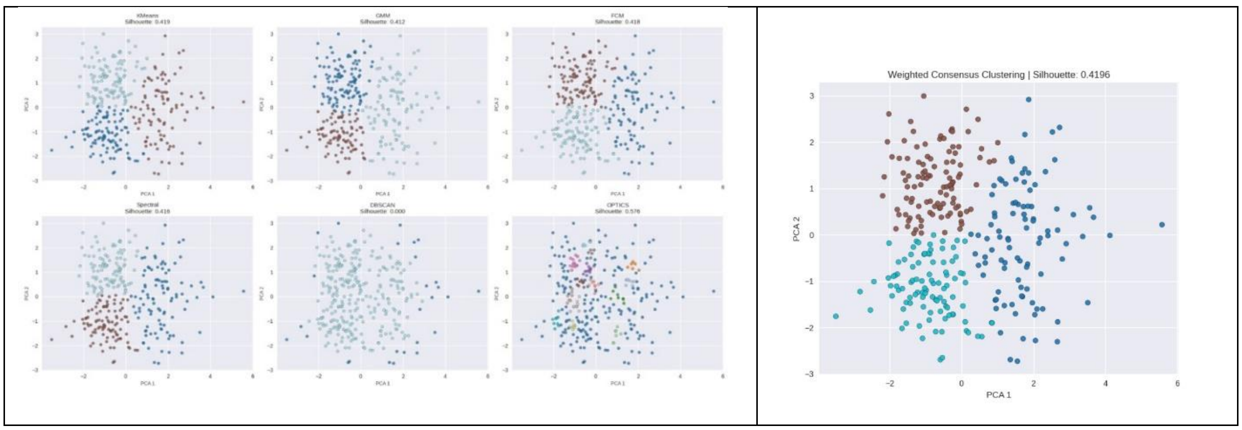

Fig 6: Cluster visualization for *Heart_failure_clinical_records_dataset.csv* dataset

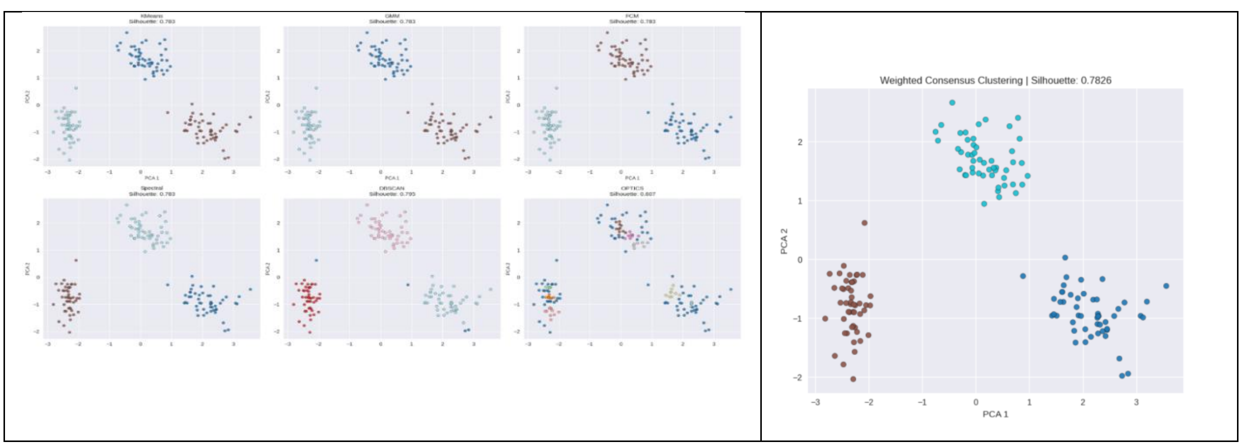

Fig 7: Cluster visualization for *IRIS.csv* dataset

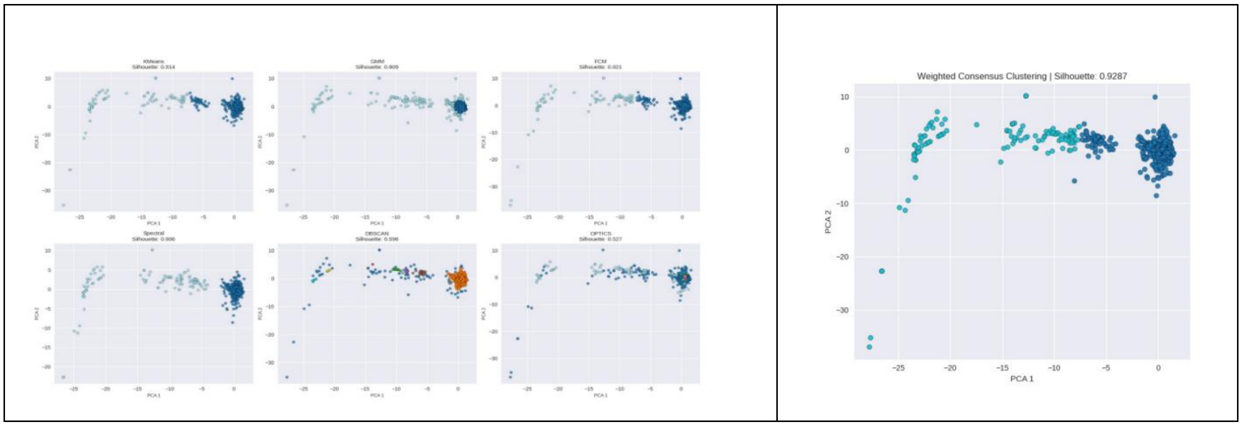

Fig 8: Cluster visualization for *creditcardfraud.csv* dataset

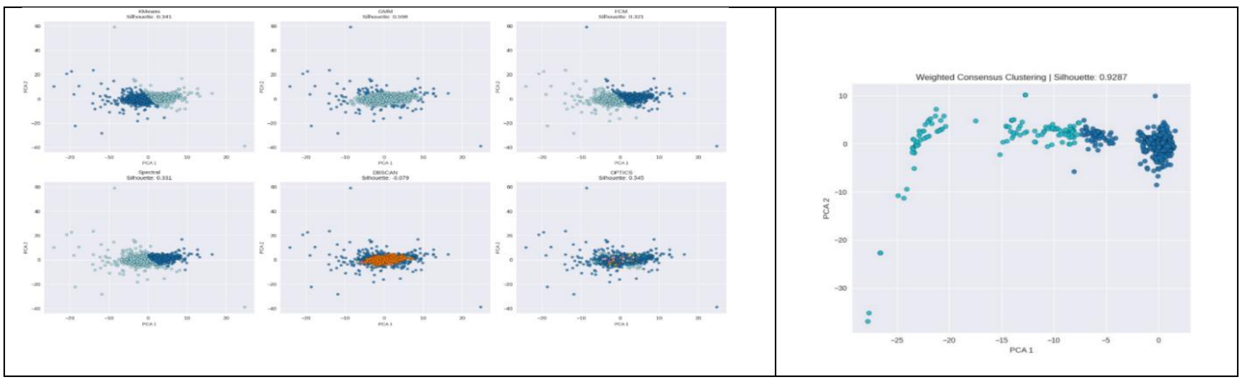

Fig 9: Cluster visualization for *movies.csv* dataset

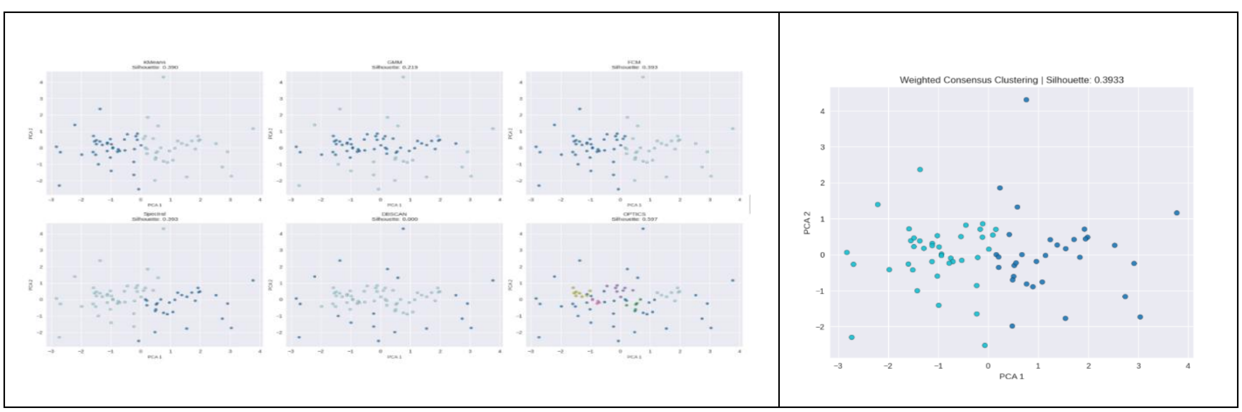

Fig 10: Cluster visualization for *accidental-deaths-in-usa-monthly.csv* dataset

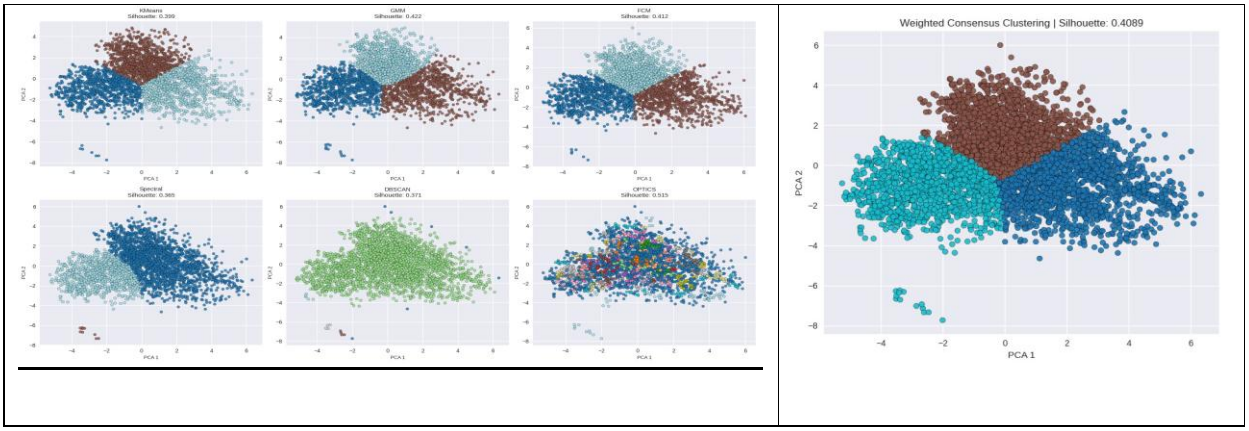

Fig 11: Cluster Visualization for *hour.csv* dataset

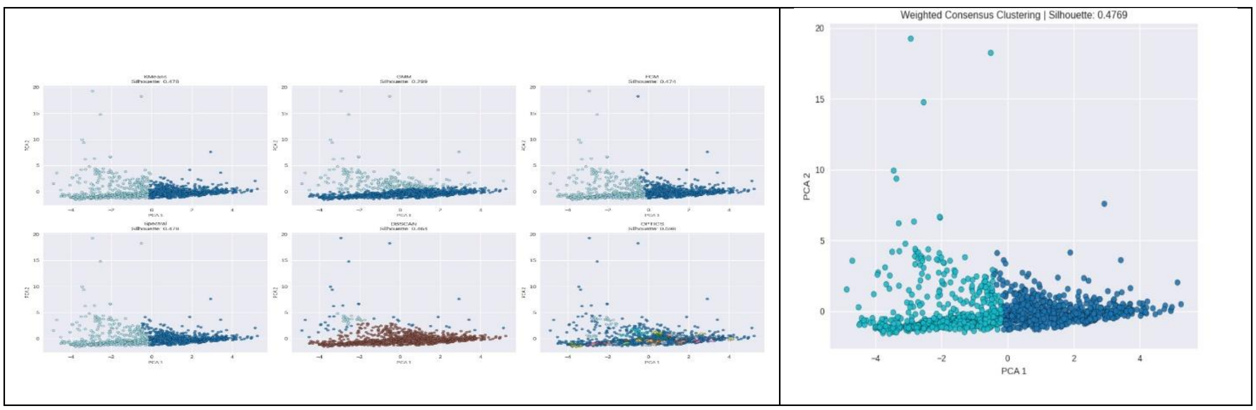

Fig 12: Cluster Visualization for *Delhi_temp.csv* dataset

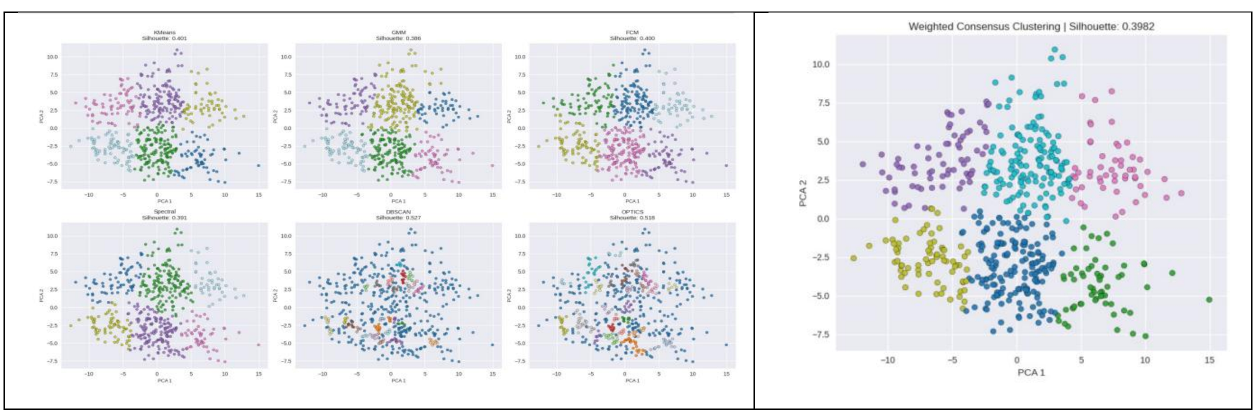

Fig 13: Cluster visualization for *Cortex_data.csv* dataset

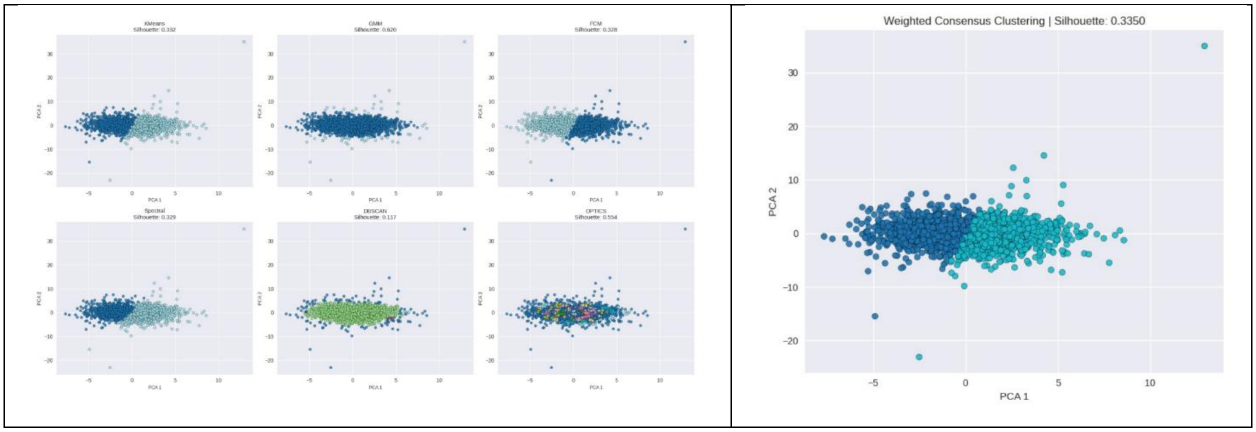

Fig 14: Cluster visualization for *Bank_transactions_data_2.csv* dataset

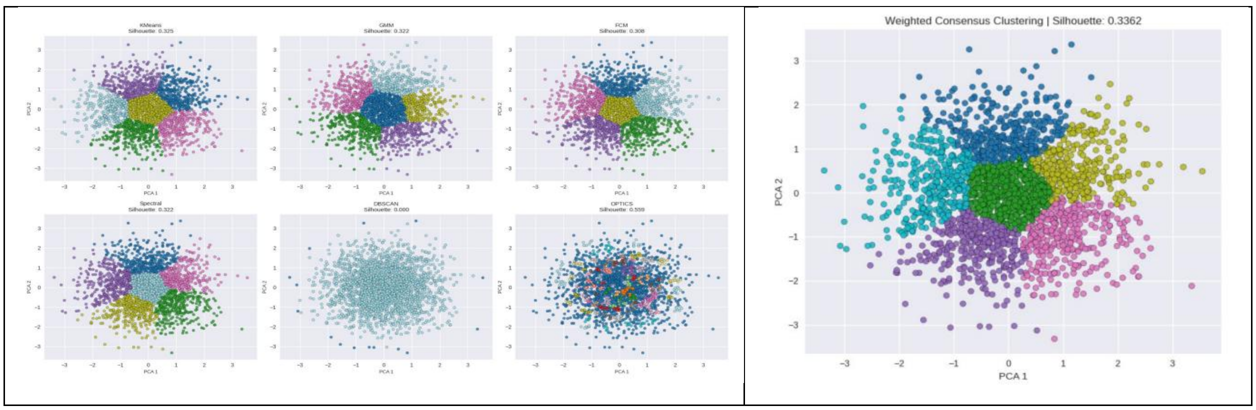

Fig 15: Cluster visualization for *marketing_campaign.csv* dataset

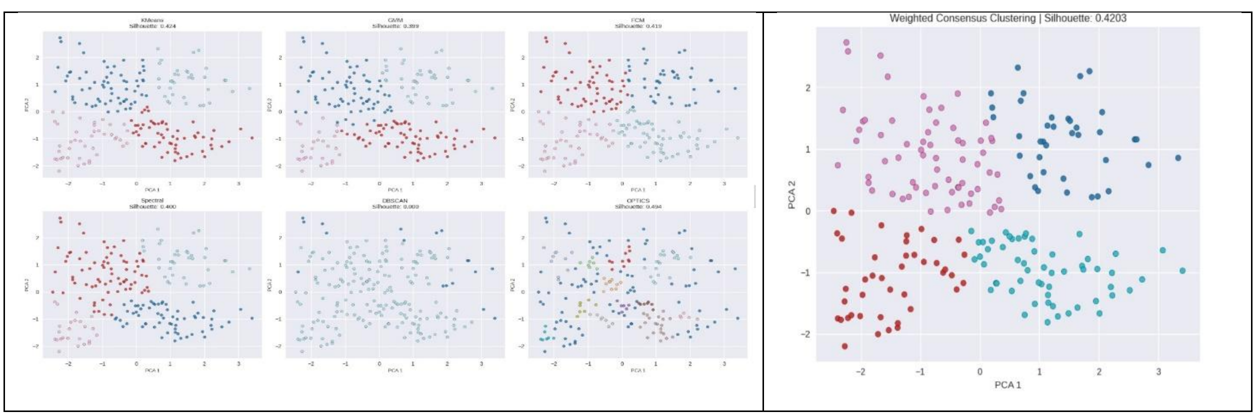

Fig 16: Cluster visualization for *Mall_Customers.csv* dataset

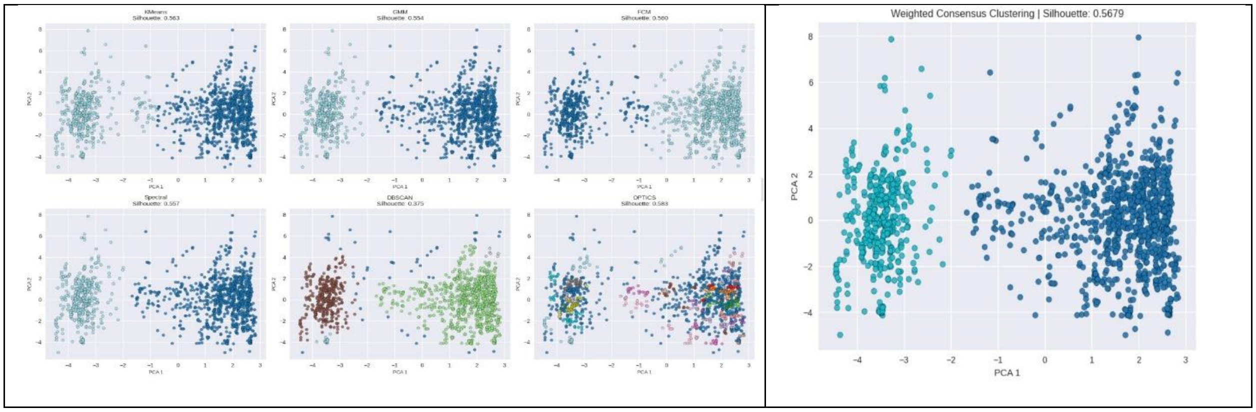

Fig 17: Cluster visualization for *Cosmetics.csv* dataset

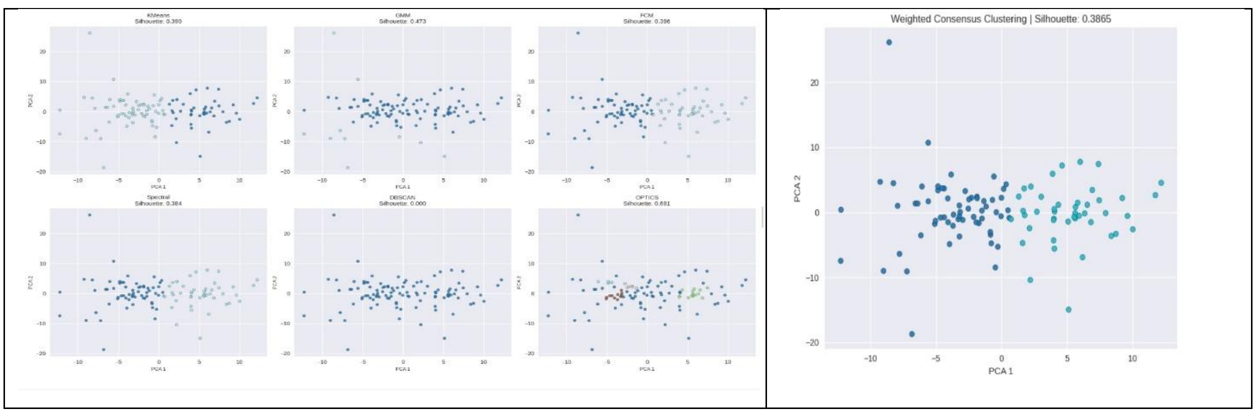

Fig 18: Cluster visualization for *World-data-2023.csv* dataset

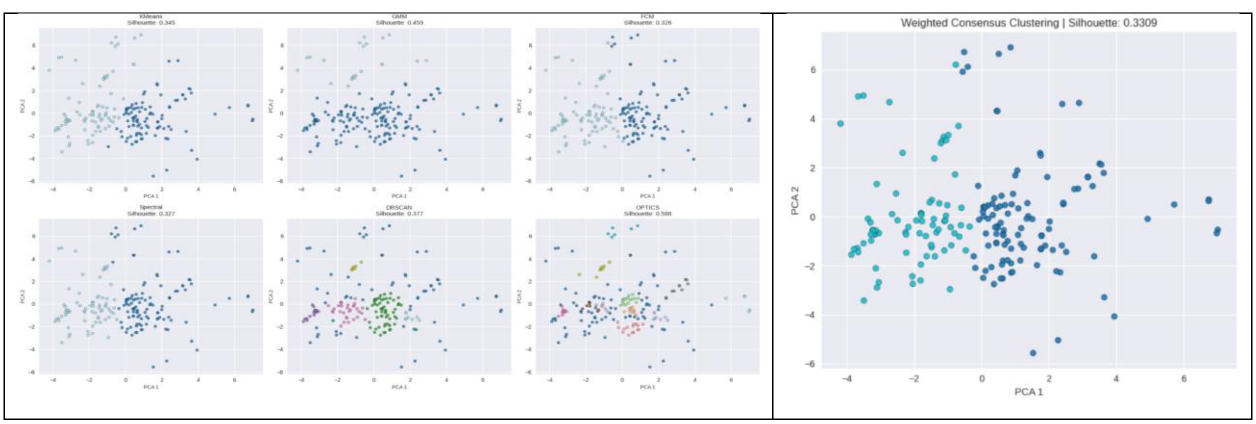

Fig 19: Cluster visualization for *Titanic.csv* dataset

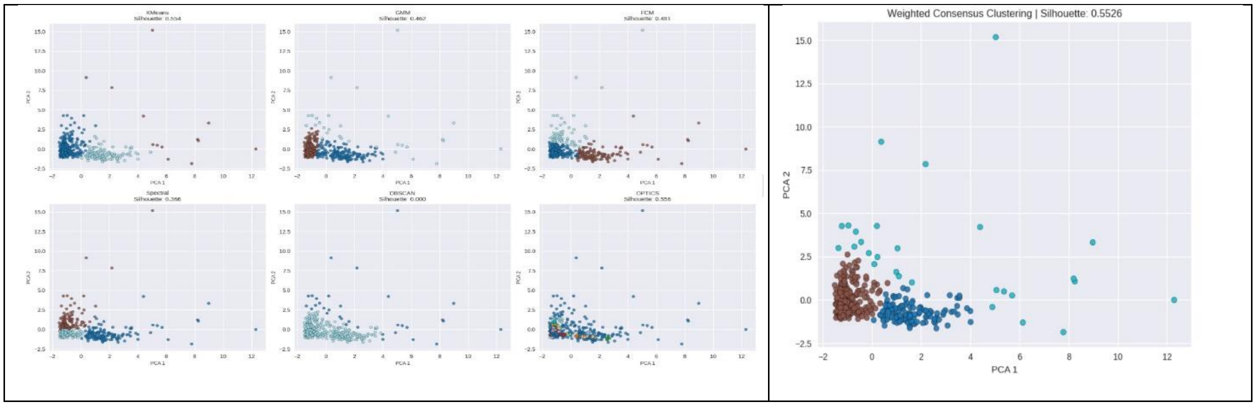

Fig 20: Cluster visualization for *customers.csv* dataset

We have done a comparison study [Table 2] with the work done by Li et al. (Li and Ding 2008) based on **accuracy score** for several datasets. We have also shown our model performance on those datasets.

Name of Dataset	**Existing model** (Li and Ding)	**Our model**
Wine	0.72	0.9663
Glass	0.49	0.5083
Zoo	0.70	0.7647
Soyabean	0.91	0.9477
Iris	0.89	0.9723

Table 2: Comparison study of our proposed algorithm with existing model (Li and Ding)

So, from the result tables [Table 1, 2] and the clustering visualization graphs [Figure 1 – 10], we can see that in most of the times we have seen improvement in clustering quality compared to any one of the tradition clustering techniques and some of the already existing weighted consensus clustering algorithms.

Chapter 7

Conclusion

In this work, we introduced a **Weighted Consensus Clustering (WCC)** algorithm that builds upon traditional consensus clustering by incorporating clustering validity metrics to guide the ensemble process. Unlike standard methods that treat all clustering outputs equally, our approach assigns **weights based on the Silhouette Score**, allowing more reliable clustering results to have a greater influence on the final consensus. This not only improves clustering accuracy but also enhances stability, especially when dealing with noisy or complex datasets.

The WCC algorithm is flexible and can be applied across various domains where unsupervised learning is essential—such as customer segmentation, anomaly detection, and pattern discovery in unlabelled data. Its performance advantages make it particularly suitable for tasks requiring robust clustering solutions.

Looking ahead, there are several promising directions for further research. One is to explore alternative cluster validity metrics, such as the **Davies-Bouldin Index**, which may provide complementary insights for weighting. Another is to adapt the method for **streaming data**, where clusters need to evolve in real-time. Additionally, applying WCC to **high-dimensional data** like biomedical or text datasets could test its scalability and generalizability. Combining WCC with **supervised feedback** in a hybrid ensemble framework could further refine results and unlock new use cases.

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